

INE4106 — Business Process Reengineering

Midterm Study Pack · 19 August, 10:30 · 40 Multiple-Choice Questions · Lecture Notes 1–10

Scope. Lecture Notes 1–10: BPR · the Hammer paper · Change Management · Introduction to ERP · Introduction to SAP · SAP Concepts · SAP S/4HANA · SD · MM · PP. Notes 11–18 (WM, FI, CO, HCM, OM, Competitiveness, Process Selection, Product Design) are **out of scope**, as are the Kaplan & Norton, Linear Programming and NSPE Ethics readings.

Format. 40 multiple-choice questions, no calculation. That means the exam is a **terminology and distinction** test: definitions, which unit sits above which, which principle belongs to whom, and which number goes with which case. Precision beats general understanding here.

Where marks are actually lost. Client vs Company Code · the three components of a Sales Area · Purchasing Organization vs Purchasing Group · master data vs transaction data · Hammer's seven principles vs the lecture's five · Kotter's eight steps vs the seven-step change process · the Ford and MBL numbers.

Part 1 — The Ten Notes at a Glance

Note	Core content	Most examinable point
1. BPR	Definition, purpose, 5 principles, 5 implementation steps, IT as enabler, benefits, risks, BPR vs continuous improvement	BPR is radical ; Kaizen/TQM are incremental . Target is a 50–70 per cent leap.
2. Hammer paper (HBR 1990)	"Don't Automate, Obliterate", the seven principles of reengineering, Ford, Mutual Benefit Life, Hewlett-Packard	The seven principles and the Ford/MBL numbers.
3. Change Management	Definition, objectives, 5 types of change, 7-step process, Kotter's 8 steps, resistance, leadership	Kotter 1996, eight steps, starting with urgency and ending with anchoring in the culture .
4. Introduction to ERP	Single software + common database, MRP to MRP II to ERP, ERP claims, conception vs reality, advantages and disadvantages, module list	Every advantage has a paired disadvantage; vendors sell modules , not whole systems.
5. Introduction to SAP	SAP company facts, Business Suite, NetWeaver layers, Enhancement Packages, the three data types	Organizational / Master / Transaction data and what belongs in each.
6. SAP Concepts	Organizational units in logistics, procurement, sales, accounting and HCM; master data; process data; documents; document flow	The hierarchies, and Sales Area = Sales Org + Distribution Channel + Division .
7. SAP S/4HANA	Intelligent Enterprise, company history, R/2 to R/3 to ERP to S/4HANA, Fiori, data types repeated	The timeline dates and that S/4HANA means Suite for HANA .
8. SD (Sell)	SD org structure incl. Credit Control Area and Sales Area, customer master, condition master, output, Order-to-Cash	Customer master is created by Sales Area and split into General / Company Code / Sales Area data.
9. MM (Buy)	Plant, Storage Location, Purchasing Organization, Purchasing Group, vendor master, info record, Procure-to-Pay	Vendor master views: General (client), Company Code (FI), Purchasing Org (MM).
10. PP (Make)	Work Center, BOM, Routing, Product Group, material planning, production planning, manufacturing execution	Work Center is an organizational unit but stored as master data ; BOM single- vs multi-level.

1.1 The integrated business process — the diagram everything hangs on

Sell (SD): receive customer inquiry -> create and send quotation -> receive customer purchase order -> create sales order -> *sufficient finished goods?*

Yes -> warehouse prepares shipment (pick and pack) -> sends shipment -> accounting creates and sends invoice -> receives payment.

No -> Make (PP): warehouse requests production -> production authorises it -> *sufficient raw materials?*

Yes -> issue raw materials -> create product -> receive finished goods.

No -> Buy (MM): create requisition -> purchasing creates and sends the purchase order -> warehouse receives shipment; in parallel accounting receives the invoice and sends payment.

Document flow in the system: Inquiry -> Quotation -> Standard Order -> Outbound Delivery -> Picking request -> Goods issue -> Invoice -> Accounting document. The status is revised every time any document in the chain changes.

1.2 Organizational hierarchies — learn these cold

Area	Hierarchy	Note
Logistics	Client > Company Code > Plant > Storage Location	Client is the highest unit and an independent environment.
Sales (SD)	Company Code > Sales Organization > Distribution Channel > Division	Sales Area = Sales Org + Distribution Channel + Division , and it determines pricing conditions.

Area	Hierarchy	Note
Procurement (MM)	Purchasing Organization can be central (above several company codes), cross-plant , or plant-specific	Purchasing Group is not above or below it — it is the buyer or group of buyers, the channel of communication to vendors.
Accounting	Client > Controlling Area > Chart of Accounts > Company Codes	One controlling area can span several company codes.
HCM	Client > Company Code > Personnel Area > Personnel Subarea	Two levels below the company code.
Credit (SD)	Credit Control Area may include one or more company codes	Grants and monitors the customer credit limit.

Part 2 — The Numbers, Names and Dates

A 40-question multiple-choice paper will put several of these straight into the options. They are cheap marks if you have them and impossible to reason out if you do not.

2.1 The three cases in the Hammer paper

Case	Before	After	The rule that changed
Ford accounts payable	500+ people in North America; matched 14 data items between receipt record, purchase order and invoice; the modest goal was a 20 per cent head-count cut	75 per cent reduction in head count; matching only 3 items — part number, unit of measure, supplier code; invoiceless processing	"We pay when we receive the invoice " became "We pay when we receive the goods ." Mazda ran the same function with 5 people.
Mutual Benefit Life insurance applications	Up to 30 discrete steps , 5 departments , 19 people ; best case 24 hours, typical 5 to 25 days ; another insurer measured 22 days in process against 17 minutes of actual work	The case manager owns the application end to end; now as little as 4 hours , average 2 to 5 days; 100 field office positions eliminated; more than twice the volume handled	The president demanded a 60 per cent improvement, which was too big to reach by tinkering.
Hewlett-Packard purchasing	More than 50 manufacturing units each with its own purchasing department; responsive but no scale, poor quantity discounts	150 per cent improvement in on-time deliveries, 50 per cent reduction in lead times, 75 per cent reduction in failure rates, lower cost of goods	Divisional purchasing kept, plus a corporate unit and a shared vendor database — treat dispersed resources as though centralized.

2.2 SAP company and product timeline

Year	Event	Year	Event
1972	SAP founded in Walldorf, Germany ; real-time data application software	2002	30th birthday; third-largest independent software provider
1977	Oracle founded (first commercial SQL in 1979)	2004	SAP ERP
1979	R/2 generation begins (introduced 1981/82)	2005/06	Release of SAP ERP announced
1986–89	R/3 development; SAP presents at CeBIT Hanover	2009	Business Suite 7
1992	R/3	2011	First in-memory product: the SAP HANA platform ; data access in seconds
1993/94	Partnership with Microsoft (R/3 on Windows NT); IBM adopts R/3	2013	Business Suite moves to HANA — fastest growing product in the history of enterprise software
1995/96	SAP joins the Internet; R/3 usable online	2015	SAP S/4HANA — "Suite for HANA", the biggest innovation since R/3

Other SAP figures from the slides: world's largest business software company and third-largest independent software provider · over **65,000** employees in more than **75** countries · **232,000+** customers in more than **180** countries · **200,000+** installations · **2,500+** business partners · total revenue more than **EUR 30 billion** a year. The S/4HANA deck quotes **400,000** customers in **190** countries. Course system: **S/4HANA 2022** with **Fiori 3.0**, curriculum **Global Bike 4.2**.

2.3 Reasons companies implement ERP (ranked)

Rank	Reason	Share	Rank	Reason	Share
1	Replace legacy system	16%	7	Reduce working capital	8%
2	Position company for growth	13%	8	Appease parent company / stakeholders	7%
3	Improve business performance	13%	9	Better serve customers	5%
4	Reporting / regulatory compliance	10%	10	Integrate systems across locations	5%
5	Make employees' jobs easier	10%	11	Other companies have ERP	2%
6	Standardize global operations	10%	12	Other	1%

2.4 Lists you may be asked to complete

List	Members
Hammer's seven principles of reengineering	1. Organize around outcomes, not tasks · 2. Have those who use the output of the process perform the process · 3. Subsume information-processing work into the real work that produces the information · 4. Treat geographically dispersed resources as though they were centralized · 5. Link parallel activities instead of integrating their results · 6. Put the decision point where the work is performed , and build control into the process · 7. Capture information once and at the source

List	Members
The lecture's five key principles of BPR	Focus on processes not tasks · IT as an enabler · dramatic improvement not marginal change (50–70 per cent) · customer-oriented redesign · cross-functional thinking
The five steps of BPR implementation	Identify processes to be redesigned · understand and analyse current processes (As-Is) · develop a new process design (To-Be) · implement the new process · evaluate and monitor results
Kotter's eight steps (1996)	Create a sense of urgency · form a guiding coalition · develop a vision and strategy · communicate the vision · empower broad-based action · generate short-term wins · consolidate gains and produce more change · anchor new approaches in the culture
The seven -step change management process	Identify the need · plan · communicate · implement · manage resistance · monitor and evaluate · reinforce and sustain
Five types of organizational change	Strategic · structural · technological · people-centered · process-oriented
Causes of resistance to change	Fear of the unknown · loss of control or job security · lack of trust in leadership · poor communication · insufficient training
SAP NetWeaver integration layers	People Integration · Information Integration · Process Integration · Application Platform
SAP ERP modules and their one-word tags	MM "Buy" · PP "Make" · SD "Sell" · FI and CO "Track" · HCM "People" · WM "Store" · EAM "Maintain" · CS "Service"
Material master views	Basic · Sales · Purchasing · Material Planning · Forecasting · Storage · Quality · Controlling · Accounting
PP master data	Material · Bill of Materials (BOM) · Routing · Work Center · Product Group

Part 3 — 100 True / False Statements

#	Statement	Ans	Why
Note 1 — Business Process Reengineering			
1	BPR is the fundamental rethinking and radical redesign of business processes.	T	Targets cost, quality, service and speed.
2	BPR aims at small, incremental improvements.	F	That is continuous improvement; BPR is radical.
3	The term was popularised by Michael Hammer in a 1990 paper.	T	"Reengineering Work: Don't Automate, Obliterate".
4	Hammer and Champy's 1993 book is <i>Reengineering the Corporation</i> .	T	The follow-up to the HBR paper.
5	Kaizen and TQM are examples of radical transformation.	F	They are continuous, incremental improvement.
6	BPR and continuous improvement can be combined in one organization.	T	BPR for transformation, CI for refinement.
7	BPR focuses on individual tasks and departments.	F	On processes that produce customer value.
8	Information technology is treated as an enabler of BPR.	T	ERP, workflow automation, analytics, cloud.
9	A typical BPR performance target is a 50 to 70 per cent improvement.	T	Not a few percentage points.
10	Every process redesign should start from the customer's needs.	T	Customer-oriented redesign.
11	Documenting the existing process is called the To-Be analysis.	F	The existing process is As-Is .
12	The final step of BPR implementation is to evaluate and monitor results.	T	Adjust further if needed.
13	Employee resistance is listed as a challenge of BPR.	T	Along with cost, disruption and weak commitment.
14	Ford's accounts payable redesign is the lecture's headline BPR example.	T	75 per cent staff reduction.
Note 2 — The Hammer Paper			
15	Hammer argues that heavy IT investment alone had delivered disappointing results.	T	Companies mechanised old ways of doing business.
16	"Stop paving the cow paths" means stop embedding outdated processes in software.	T	Obliterate rather than automate.
17	Reengineering can be planned meticulously and accomplished in small cautious steps.	F	Hammer calls it an all-or-nothing proposition.
18	"Organize around outcomes, not tasks" is one of Hammer's principles.	T	One person performs all the steps.
19	"Have those who use the output of the process perform the process" is a Hammer principle.	T	Departments make their own purchases.
20	"Capture information once and at the source" is a Hammer principle.	T	Bar coding, relational databases, EDI.
21	"Integrate parallel activities instead of linking them" is a Hammer principle.	F	Reversed: link parallel activities instead of integrating their results.
22	"Put the decision point where the work is performed" flattens the organization.	T	Build control into the process.
23	Ford originally aimed to cut accounts payable head count by about 20 per cent.	T	It achieved 75 per cent by reengineering.
24	Mazda's accounts payable organization consisted of five people.	T	Against more than 500 at Ford North America.
25	Under Ford's new process the system matches fourteen data items.	F	Three: part number, unit of measure, supplier code.
26	Ford's old rule was "We pay when we receive the goods".	F	Old: when we receive the invoice . New: the goods.
27	Mutual Benefit Life created a role called the case manager.	T	Total responsibility from receipt to issue.
28	MBL's old process involved as many as 30 steps across five departments.	T	And 19 people.
29	MBL's president demanded a 60 per cent improvement in customer service.	T	Too ambitious for tinkering.
30	After reengineering, MBL eliminated 100 field office positions.	T	And handled more than twice the volume.
31	Hewlett-Packard centralised purchasing completely into one department.	F	It kept divisional units and added a corporate coordinator with a shared database.
32	HP reported a 150 per cent improvement in on-time deliveries.	T	Plus 50 per cent shorter lead times and 75 per cent fewer failures.

#	Statement	Ans	Why
33	Reengineering affects only the business process itself.	F	Job designs, structures and management systems must all be refashioned.
34	Executive leadership with real vision is necessary for reengineering to succeed.	T	Otherwise cynics outlast the effort.
Note 3 — Change Management			
35	Change management covers both the technical and the human side of change.	T	Systems and structures; attitudes and behaviours.
36	Minimising resistance and increasing engagement is an objective of change management.	T	One of the stated goals.
37	Technological change means adopting new tools, automation or digital systems.	T	One of the five types.
38	Redesigning workflows and procedures is called people-centered change.	F	That is process-oriented change.
39	Communicating the change comes before implementing it.	T	Step 3 precedes step 4.
40	Resistance is treated as an abnormal reaction that indicates a failing project.	F	It is described as a natural reaction.
41	Kotter's model was proposed in 1996 and has eight steps.	T	It emphasises leadership and communication.
42	Kotter's first step is to form a guiding coalition.	F	First is to create a sense of urgency .
43	Kotter's final step is to anchor new approaches in the culture.	T	Prevents backsliding.
44	Generating short-term wins is one of Kotter's eight steps.	T	Between empowerment and consolidation.
45	Lack of trust in leadership is listed as a cause of resistance.	T	With fear of the unknown and poor communication.
46	Change management is a one-time event completed at go-live.	F	A continuous process of adaptation.
Notes 4, 5 and 7 — ERP, SAP and S/4HANA			
47	ERP runs the company from a single software package with a common database.	T	Integrated business environment.
48	ERP evolved from MRP and MRP II and typically retains an MRP core.	T	Beginning in the 1970s.
49	SAP was founded in 1972 in Walldorf, Germany.	T	Standard software for integrated business solutions.
50	Oracle was founded in 1977 and offered the first commercial SQL in 1979.	T	Stated on the ERP slide.
51	ERP systems were among the largest IT investments of the 1990s.	T	Attention grew from the 1990s onward.
52	In practice ERP vendors usually sell complete systems rather than modules.	F	They would like to; buyers start smaller scale to cut cost and risk.
53	Greater data integration makes errors easier to correct.	F	The slide says integration makes data harder to correct .
54	System integration is listed as bringing less flexibility.	T	Every advantage has a paired disadvantage.
55	Training costs for ERP are typically underbudgeted.	T	Along with other hidden implementation costs.
56	SAP is described as the world leader in enterprise software by total revenue.	T	More than EUR 30 billion a year.
57	SAP ERP is designed to be multi-lingual, multi-currency and multi-balance.	T	Parallel general-ledger accounting.
58	The SAP Business Suite consists of ERP surrounded by PLM, SRM, CRM and SCM on NetWeaver.	T	ERP is the centre.
59	SAP NetWeaver has four integration layers, one of which is Process Integration.	T	With People, Information and Application Platform.
60	SAP Enhancement Packages are a kind of support package.	F	Explicitly not support packages; they are optional innovations.
61	S/4HANA stands for "Suite for HANA" and was released in 2015.	T	Described as the biggest innovation since R/3.
62	SAP R/3 belongs to the 1992 generation and SAP ERP to 2004.	T	R/2 1979, R/3 1992, ERP 2004, S/4HANA 2015.
63	SAP's first in-memory computing product appeared in 2011.	T	The SAP HANA platform; data access in seconds.
64	The SAP Business Suite moved to HANA in 2013.	T	The fastest growing product in the history of enterprise software.
65	Fiori is the user interface used to interact with S/4HANA.	T	The course uses Fiori 3.0.
Note 6 — SAP Concepts			

#	Statement	Ans	Why
66	The client is the highest organizational unit in SAP.	T	An independent environment representing the enterprise.
67	The company code is the smallest unit for which a legal set of books can be maintained.	T	The standard definition.
68	A plant corresponds to a warehouse in SAP terminology.	F	A plant is a factory; a storage location is the warehouse.
69	Storage locations sit below plants.	T	Client > Company Code > Plant > Storage Location.
70	A purchasing organization must always belong to exactly one plant.	F	It may be central, cross-plant or plant-specific.
71	A sales area is the combination of sales organization, distribution channel and division.	T	It determines pricing conditions.
72	A division is a way of grouping materials, products or services.	T	Bicycles, accessories and so on.
73	A controlling area may contain several company codes.	T	Client > Controlling Area > Chart of Accounts > Company Codes.
74	Personnel subareas sit directly under the company code.	F	Under personnel areas , which sit under the company code.
75	Master data is static data stored for a long time and seldom changed.	T	Customer, material, vendor, conditions.
76	Material master data is stored in functional segments called views.	T	Basic, sales, purchasing, planning and so on.
77	Stock quantity in the material master is held at client level.	F	At storage location level; name and weight are at client level.
78	Process data is dynamic data created during a business process.	T	Contains organizational, master and situational data.
79	A document is generated data that represents a business process in the system.	T	Sales order, purchase order, material document, financial document.
80	The document flow lets you see the status of an order at any point in time.	T	SAP revises the status on every change.
Notes 8, 9 and 10 — SD, MM and PP			
81	The credit control area grants and monitors customer credit limits.	T	It may include one or more company codes.
82	Wholesale, retail and direct sales are typical distribution channels.	T	The way products reach the customer.
83	Shipping point, loading point, sales office, sales group and sales person are SD organizational units.	T	Beyond the core sales area units.
84	Every customer must have a customer master record.	T	Stated explicitly in the SD deck.
85	The customer master is created at plant level.	F	By sales area .
86	Customer master data is divided into general data, company code data and sales area data.	T	Three areas.
87	Condition master data covers prices, surcharges, discounts, freights and taxes.	T	Can be material- or customer-specific.
88	Output can be sent by e-mail, mail or EDI.	T	Quotation, confirmation, invoice.
89	The SD process taught in the course is Procure-to-Pay.	F	SD is Order-to-Cash ; MM is Procure-to-Pay.
90	A plant may be a manufacturing, distribution, purchasing or maintenance facility.	T	An operating area or branch.
91	The purchasing organization negotiates purchase conditions with vendors.	T	It performs the buying activity for a plant.
92	The purchasing group is an organizational level above the purchasing organization.	F	It is the buyer or group of buyers, and the channel of communication to vendors.
93	Every vendor must have a vendor master record.	T	Maintained by Purchasing and Accounting.
94	Vendor master company code data includes the reconciliation account and terms of payment.	T	The FI view.
95	Purchasing currency and vendor partners sit in the purchasing organization view.	T	The MM view.
96	A work center defines where and when an operation is performed.	T	It has an available capacity.
97	Work center activities are valued by charge rates from cost centers and activity types.	T	Links PP to CO.
98	A work center can be a machine, a person, a production line or a group of craftsmen.	T	Not only equipment.

#	Statement	Ans	Why
99	A bill of materials lists the components that make up a product or assembly.	T	Single-level or multi-level.
100	A multi-level BOM shows only the immediate components of the finished product.	F	That is single-level ; multi-level explodes sub-assemblies.
101	Routing and product group are PP master data.	T	With material, BOM and work center.

Part 4 — 60 Multiple Choice Questions

#	Question and options	Ans	Why
1	BPR is best defined as: A) Gradual refinement of existing procedures B) Fundamental rethinking and radical redesign of business processes C) Automation of existing workflows D) Outsourcing of non-core functions	B	For dramatic improvement in cost, quality, service and speed.
2	Which pair are examples of continuous improvement rather than BPR? A) ERP and workflow automation B) Kaizen and TQM C) Ford and MBL D) As-Is and To-Be	B	Incremental, not radical.
3	The title of Hammer's 1990 HBR article is: A) Reengineering the Corporation B) Reengineering Work: Don't Automate, Obliterate C) The Balanced Scorecard D) Integrated Business Processes with ERP Systems	B	The 1993 book is <i>Reengineering the Corporation</i> .
4	Documenting how the existing process works is called the: A) To-Be process B) As-Is process C) Requisite process D) Benchmark process	B	Step 2 of BPR implementation.
5	In BPR, information technology should be used to: A) Speed up the existing process B) Enable a new process design C) Replace the workforce D) Document the As-Is state	B	Not to automate an existing one.
6	Which is NOT one of Hammer's seven principles? A) Organize around outcomes, not tasks B) Capture information once and at the source C) Automate every manual step before redesigning D) Put the decision point where the work is performed	C	That is the opposite of Hammer's argument.
7	"Treat geographically dispersed resources as though they were centralized" is illustrated by: A) Ford B) Mutual Benefit Life C) Hewlett-Packard D) Mazda	C	Divisional purchasing plus a corporate unit and shared database.
8	Ford's reengineered accounts payable process matches how many data items? A) 14 B) 3 C) 5 D) 30	B	Part number, unit of measure and supplier code.
9	Ford's head-count reduction in accounts payable was: A) 20 per cent B) 50 per cent C) 75 per cent D) 100 per cent	C	Against an original goal of about 20 per cent.
10	The old rule that Ford's reengineering replaced was: A) "We pay when we receive the invoice" B) "We pay when we receive the goods" C) "Local inventory is needed for good service" D) "Credit decisions are made by the credit department"	A	It became "We pay when we receive the goods."
11	The new role created at Mutual Benefit Life was the: A) Process owner B) Case manager C) Senior underwriter D) Field supervisor	B	Total responsibility from receipt to issue.
12	MBL's old application process spanned: A) 3 steps and 2 departments B) As many as 30 steps, 5 departments and 19 people C) 17 steps and 4 departments D) 100 steps and 10 departments	B	Typical turnaround was 5 to 25 days.

#	Question and options	Ans	Why
13	Hammer describes reengineering as: A) A carefully phased, low-risk programme B) An all-or-nothing proposition with an uncertain result C) A purely technical exercise D) A synonym for downsizing	B	It cannot be done in small cautious steps.
14	Change management is best described as covering: A) Only systems and structures B) Only employee attitudes C) Both the technical and the human side of change D) Only the go-live period	C	Processes and systems plus attitudes and behaviours.
15	Which is NOT one of the five types of organizational change listed? A) Strategic B) Structural C) Financial D) People-centered	C	The fifth is process-oriented.
16	Kotter's eight-step model begins with: A) Forming a guiding coalition B) Creating a sense of urgency C) Developing a vision D) Communicating the vision	B	Urgency first, culture last.
17	Kotter's model was published in: A) 1990 B) 1993 C) 1996 D) 2015	C	Hammer 1990, Hammer and Champy 1993.
18	Which is the final step of Kotter's model? A) Generate short-term wins B) Consolidate gains C) Anchor new approaches in the culture D) Empower broad-based action	C	Prevents regression to old habits.
19	Which is listed as a cause of resistance to change? A) Excess training B) Fear of the unknown C) Too much communication D) Excessive delegation	B	With loss of control, distrust, poor communication and insufficient training.
20	ERP evolved from: A) CRM and SRM B) MRP and MRP II C) SQL and relational databases D) TQM and Kaizen	B	ERP typically retains an MRP core.
21	Which statement about ERP in practice is correct? A) Vendors normally sell only the complete integrated system B) Vendors would like to sell full systems but usually sell modules C) Buyers always convert the entire system at once D) Modules cannot be bought separately	B	Buyers reduce cost and risk by starting at smaller scale.
22	Which is a disadvantage paired with data integration? A) Less flexibility B) Errors are harder to correct C) Less freedom and creativity D) Higher training costs	B	System integration gives less flexibility; better methods give less freedom.
23	The most frequently cited reason for implementing an ERP system is: A) Replace legacy system B) Position company for growth C) Improve business performance D) Better serve customers	A	16 per cent, ahead of growth and performance at 13 per cent each.
24	In the module shorthand used in the course, "Make" is: A) MM B) SD C) PP D) WM	C	MM buys, SD sells, WM stores.
25	SAP was founded in: A) 1970 in Munich B) 1972 in Walldorf C) 1977 in California D) 1992 in Hanover	B	Oracle was founded in 1977.

#	Question and options	Ans	Why
26	Which four layers make up SAP NetWeaver? A) Sales, Purchasing, Production, Finance B) People, Information, Process Integration and Application Platform C) Client, Company Code, Plant, Storage Location D) PLM, SRM, CRM, SCM	B	The last option lists the Business Suite components around ERP.
27	SAP Enhancement Packages are: A) Mandatory support packages B) Optionally installed and activated software innovations C) Database upgrades D) Third-party add-ons	B	Explicitly not support packages.
28	S/4HANA stands for: A) Systems for HANA B) Suite for HANA C) Standard for HANA D) Solution for HANA	B	Released 2015, the biggest innovation since R/3.
29	Which sequence of SAP generations is correct? A) R/3, R/2, ERP, S/4HANA B) R/2, R/3, ERP, S/4HANA C) ERP, R/2, R/3, S/4HANA D) R/2, ERP, R/3, S/4HANA	B	1979, 1992, 2004, 2015.
30	SAP's first in-memory computing product was launched in: A) 2005 B) 2009 C) 2011 D) 2015	C	The SAP HANA platform; the Business Suite moved onto it in 2013.
31	Which is an example of organizational data? A) Customer B) Material C) Company Code D) Purchase Order	C	Customer and material are master data; a purchase order is transaction data.
32	Which is an example of transaction data? A) Vendor B) Plant C) Condition D) Sales Order	D	Vendor and condition are master data; plant is organizational data.
33	The highest organizational unit in an SAP system is the: A) Company code B) Client C) Controlling area D) Plant	B	An independent environment representing the enterprise.
34	The smallest organizational unit for which a legal set of books can be maintained is the: A) Client B) Company code C) Plant D) Business area	B	Standard definition.
35	In SAP terminology, a warehouse corresponds to a: A) Plant B) Storage location C) Shipping point D) Division	B	A factory corresponds to a plant.
36	A sales area consists of: A) Company code, plant and storage location B) Sales organization, distribution channel and division C) Sales office, sales group and sales person D) Client, company code and sales organization	B	It determines pricing conditions.
37	Which organizational unit grants and monitors a customer's credit limit? A) Sales organization B) Credit control area C) Company code D) Controlling area	B	It can include one or more company codes.
38	The accounting hierarchy taught in the course is: A) Client > Controlling Area > Chart of Accounts > Company Codes B) Client > Company Code > Controlling Area C) Company Code > Client > Chart of Accounts D) Controlling Area > Client > Company Code	A	One controlling area can span several company codes.

#	Question and options	Ans	Why
39	In human capital management, personnel subareas sit directly beneath: A) The client B) The company code C) Personnel areas D) The controlling area	C	Client > Company Code > Personnel Area > Personnel Subarea.
40	A purchasing organization may be: A) Only plant-specific B) Only central C) Central, cross-plant or plant-specific D) Only cross-company	C	All three arrangements appear on the slide.
41	The purchasing group is best described as: A) An organizational level above the purchasing organization B) The buyer or group of buyers, and the channel of communication to vendors C) A grouping of materials D) A storage location for purchased goods	B	Not a hierarchy level.
42	Which statement about the material master is correct? A) It is stored as a single flat record B) It is stored in functional segments called views C) It exists only in MM D) It contains no sales information	B	It is used by SD, MM, PP, PM, CO and QM.
43	Stock quantity in the material master is maintained at: A) Client level B) Company code level C) Storage location level D) Sales organization level	C	Name, weight and unit of measure are at client level.
44	The customer master record is created: A) At plant level B) By sales area C) At storage location level D) At controlling area level	B	Sales organization, distribution channel and division.
45	Customer master data is divided into: A) Basic, sales and storage data B) General data, company code data and sales area data C) General, purchasing and accounting data D) Client, plant and storage data	B	The third option describes the vendor master.
46	Vendor master data is divided into: A) General, sales area and plant data B) General data, company code (FI) data and purchasing organization (MM) data C) Basic, forecasting and quality data D) Client, division and channel data	B	Reconciliation account sits in the company code view.
47	Condition master data includes: A) Only list prices B) Prices, surcharges, discounts, freights and taxes C) Only customer addresses D) Only delivery dates	B	It can be material- or customer-specific.
48	The SD process taught in the course is: A) Procure-to-Pay B) Order-to-Cash C) Plan-to-Produce D) Record-to-Report	B	MM covers Procure-to-Pay.
49	The MM process taught in the course is: A) Procure-to-Pay B) Order-to-Cash C) Manufacturing execution D) Material planning	A	PP covers material planning and manufacturing execution.
50	Which sequence matches the SAP document flow? A) Quotation, Inquiry, Order, Invoice, Delivery B) Inquiry, Quotation, Order, Outbound Delivery, Goods Issue, Invoice, Accounting Document C) Order, Inquiry, Delivery, Quotation, Invoice D) Requisition, Purchase Order, Invoice, Payment	B	The last option is the procurement chain, not the sales one.

#	Question and options	Ans	Why
51	In the integrated business process, what happens when finished goods are insufficient? A) The order is rejected B) Production is requested and authorised C) The invoice is issued anyway D) The customer is asked to reorder	B	And if raw materials are also short, procurement is triggered.
52	The procurement branch of the integrated process runs: A) Purchase order, requisition, shipment, payment B) Requisition, purchase order, receive shipment, receive invoice, send payment C) Invoice, payment, requisition, shipment D) Quotation, order, delivery, invoice	B	Invoice and payment continue in parallel.
53	A work center in PP is: A) Purely organizational data with no master record B) An organizational unit that is held as master data in the system C) A type of storage location D) A financial cost object only	B	It defines where and when an operation is performed.
54	Work center activities are valued by charge rates determined by: A) Sales conditions B) Cost centers and activity types C) Purchasing info records D) Material master views	B	This is the PP link into controlling.
55	A bill of materials is: A) A list of vendors for a material B) A list of the components that make up a product or assembly C) A routing of operations D) A production schedule	B	It can be single-level or multi-level.
56	A single-level BOM shows: A) Every component down to raw materials B) Only the immediate components of the product C) Only purchased parts D) Only sub-assemblies	B	A multi-level BOM explodes the sub-assemblies as well.
57	Which set is PP master data? A) Customer, vendor, condition, output B) Material, bill of materials, routing, work center, product group C) Client, company code, plant, storage location D) Inquiry, quotation, order, invoice	B	The third option is organizational data; the fourth is documents.
58	Which storage locations appear in the Global Bike structure? A) Assembly, Packaging, Inspection B) Raw Materials, Semi-finished Goods, Finished Goods, Trading Goods, Miscellaneous C) Dallas, Miami, San Diego D) Wholesale, Internet	B	The first option lists work centers, the third plants, the fourth distribution channels.
59	In Global Bike, Wholesale and Internet are: A) Divisions B) Distribution channels C) Sales organizations D) Plants	B	Bicycles and Accessories are the divisions.
60	Which is a Global Bike company code? A) US West B) Dallas C) Global Bike Inc. D) Bicycles	C	US West is a sales organization and Dallas a plant.

Part 5 — The Pairs That Get Confused

A multiple-choice paper is built out of these. If you can separate every pair below, the distractors stop working.

This	Not this	The distinction
BPR	Continuous improvement (Kaizen, TQM)	Radical and disruptive versus incremental and gradual. BPR targets a 50–70 per cent leap.
Hammer's seven principles	The lecture's five key principles	The seven come from the 1990 HBR paper; the five are the lecture note's own summary. Know which list a question is asking for.
Kotter's eight steps	The seven-step change process	Kotter (1996) starts with urgency and ends with culture. The seven-step process starts with identifying the need and ends with reinforcing.
As-Is	To-Be	As-Is documents the current process (step 2); To-Be is the redesign (step 3).
Client	Company code	Client is the highest unit and an independent environment. The company code is the smallest unit with its own legal set of books.
Plant	Storage location	Plant = factory or branch. Storage location = warehouse, differentiating stocks of a material inside one plant.
Purchasing organization	Purchasing group	The organization procures and negotiates with vendors. The group is the buyer or buyers, a communication channel, not a hierarchy level.
Sales area	Sales organization	The sales area is the <i>combination</i> of sales organization, distribution channel and division.
Distribution channel	Division	Channel = how the product reaches the customer (wholesale, retail, internet). Division = a grouping of the products themselves.
Controlling area	Company code	One controlling area can span several company codes; the chart of accounts sits between them.
Master data	Transaction data	Master data is static and seldom changed (customer, material, vendor, condition). Transaction data records a business event (order, invoice, quotation).
Organizational data	Master data	Company code, plant, storage location, distribution channel and purchasing organization are organizational, not master, data.
Customer master split	Vendor master split	Customer: General / Company Code / Sales Area. Vendor: General / Company Code (FI) / Purchasing Organization (MM).
Order-to-Cash (SD)	Procure-to-Pay (MM)	Selling versus buying. PP covers material planning, production planning and manufacturing execution.
Single-level BOM	Multi-level BOM	Single-level lists only the immediate components; multi-level explodes the sub-assemblies too.
Work center	Storage location	The work center is where an operation is performed and has a capacity; it is an organizational unit held as master data.
Enhancement Package	Support package	Enhancement packages are optional innovations and are explicitly <i>not</i> support packages.
SAP HANA	SAP S/4HANA	HANA (2011) is the in-memory platform. S/4HANA (2015) is the business suite that runs on it.
R/3 (1992)	SAP ERP (2004)	Two different generations; R/2 is 1979 and S/4HANA 2015.
Ford	Mutual Benefit Life	Ford = accounts payable, 75 per cent cut, three matched items, invoiceless processing. MBL = insurance applications, case manager, 100 positions eliminated.

Twelve facts to check one last time before you go in.

1. BPR = fundamental rethinking, **radical** redesign; cost, quality, service, speed.
2. Hammer **1990** paper, Hammer and Champy **1993** book, Kotter **1996** model.
3. Hammer's seven principles begin with "organize around **outcomes**, not tasks" and end with "capture information **once and at the source**".
4. Ford: 500+ staff, Mazda 5, goal 20 per cent, result **75 per cent**, 14 items down to **3**.
5. MBL: 30 steps, 5 departments, 19 people; the **case manager** replaced them all.
6. Kotter starts with **urgency**, ends with **anchoring in the culture**.
7. ERP came from **MRP -> MRP II -> ERP**; SAP founded **1972 in Walldorf**.
8. Every ERP advantage has a paired disadvantage; training is **underbudgeted**.
9. Three data types: **organizational, master, transaction**.
10. **Client > Company Code > Plant > Storage Location**; **Sales Area = Sales Org + Distribution Channel + Division**.
11. Customer master is created **by sales area**; vendor master has a General, an FI and an MM view.
12. Timeline: R/2 **1979**, R/3 **1992**, ERP **2004**, HANA platform **2011**, Suite on HANA **2013**, S/4HANA **2015**.